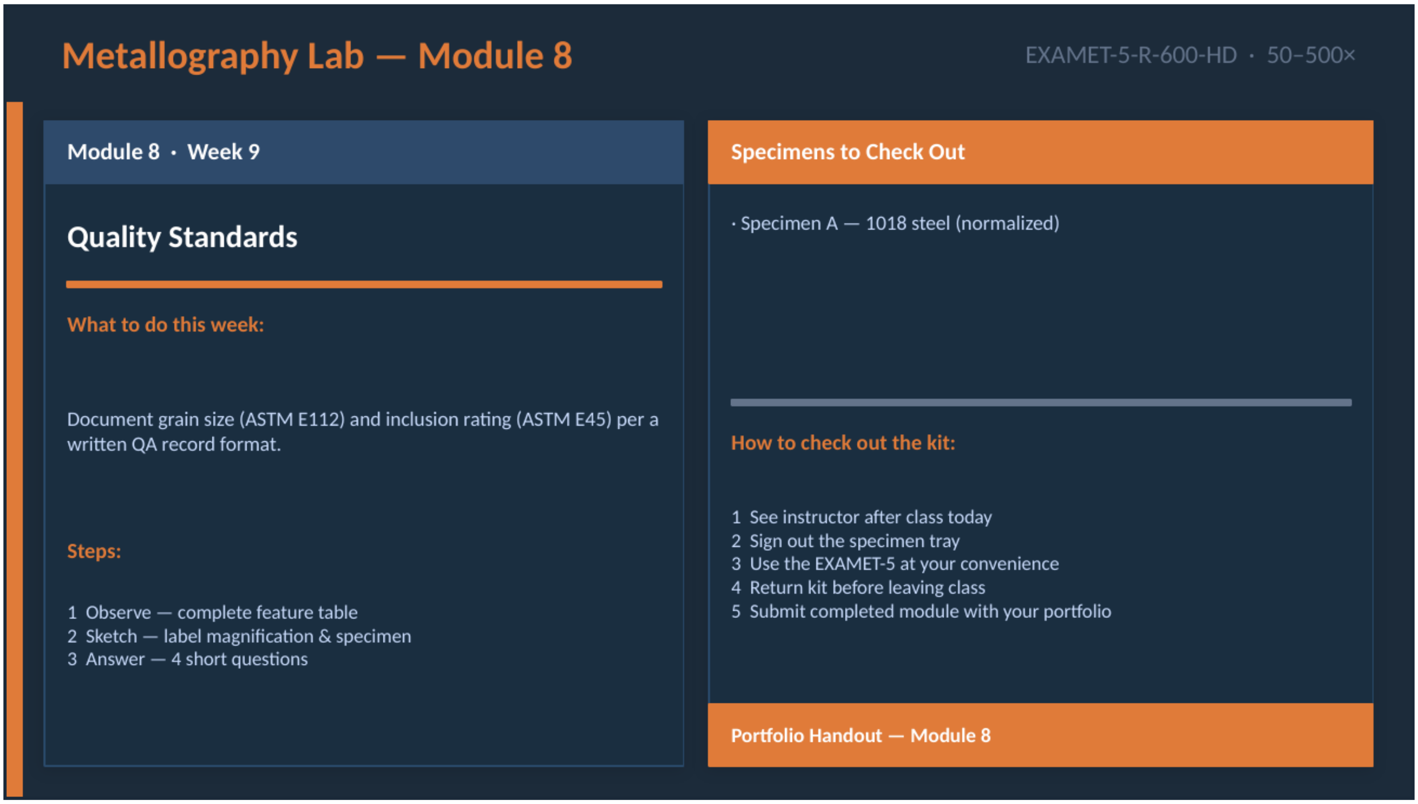
Name: ______________________ Date: ______________________

SLIDE 1

NOTES

SLIDE 2 *Class Agenda*

NOTES



SLIDE 3 *Metallography Lab — Module 8 EXAMET-5-R-600-HD · 50–500×*

NOTES

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11** Nov 2 Forming	**Wk 15** Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	**Wk 9 ← here** Oct 19 Quality Standards	**Wk 12** Nov 9 Machining	**Wk 16** Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 · FINAL EXAM

SLIDE 4 *MECH 191 — Course & Unit Roadmap*

NOTES

Week 9 — At a Glance

Topics

- Quality standards — ASTM, AWS, ASME, ISO, and API: what each organization governs and why using the wrong standard is a serious error
- Manufacturing weld defects — porosity, slag inclusions, lack of fusion, lack of penetration, cracks, undercut, and overlap
- Service defects — fatigue cracks, stress corrosion cracking, and creep damage
- Precise terminology: indication vs. discontinuity vs. defect vs. reject
- The evaluation sequence: detection → characterization → evaluation → disposition
- Inspection documentation — minimum report content, traceability, record retention, and non-conformance reports

Resources

Reference Material

Moniz Ch. 10
Materials Standards

Reference Pages

Quality Standards & Codes
Reading Inspection Documentation
Defect Recognition

Unit 3 Quiz

Covers Weeks 7–9
Scheduled this session

Next: Unit 4 — Manufacturing

SLIDE 5

NOTES

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What is a quality standard and why does it exist — what problem does it solve for a technician in the field?
2. What are the most common defects a technician will encounter and how is each one recognized during inspection?
3. What is the difference between an indication, a discontinuity, a defect, and a reject — and why does the distinction matter?
4. What does good inspection documentation look like, and why is an incomplete inspection report a serious problem?

SLIDE 6 *Core Questions*

NOTES

Quality Standards & Codes — Knowing Which Rules Apply

Major Standards Bodies

Body	Governs	Key Standard(s)	Technician Relevance
ASTM	Material specs & test methods	A36, A370, E165	Acceptance criteria for base metals; test procedures for mechanical and NDT
AWS	Welding codes & inspection	D1.1 Structural Steel	Weld acceptance criteria, inspection requirements, welder qualification
ASME	Pressure vessels, boilers & piping	BPVC Sec. V (NDT), Sec. IX	NDT methods and procedures; welding qualification for pressure equipment
ISO	International standards (global use)	9606, 17636, 17640	Welder qualification, RT and UT of welds — used on international projects
API	Oil & gas industry	1104, 570, 650	Pipeline welding, piping inspection, storage tank fabrication and repair

Using the wrong standard is as serious an error as missing a defect.
Acceptance criteria differ between codes — a weld accepted under AWS D1.1 may be rejectable under ASME Section VIII, or vice versa. Know your governing document before you inspect.

SLIDE 7 *Quality Standards & Codes — Knowing Which Rules Apply*

NOTES

Common Weld Defects — Manufacturing Discontinuities

Porosity

Cause: Gas trapped in weld pool — moisture in consumables, contaminated base metal, inadequate shielding gas, or excessive travel speed.
Appears: Rounded voids — dark circular spots on RT; round bleed-out on PT; discrete point reflectors on UT A-scan.
Risk: Reduces cross-sectional area; acts as stress concentrator. Acceptance criteria limit size, number, and spacing.

Slag Inclusions

Cause: Flux or slag from SMAW/FCAW not fully removed between passes, or trapped by poor bead geometry.
Appears: Irregular elongated dark indications on RT; irregular linear bleed-out on PT if surface-breaking; elongated UT reflectors.
Risk: Reduces cross-section; can initiate fatigue cracks. Planar slag inclusions are more serious than rounded ones.

Lack of Fusion (LOF)

Cause: Insufficient heat input, incorrect travel speed, or poor technique — weld metal does not fuse with base metal or previous pass.
Appears: Linear indication parallel to fusion face on RT (may be missed if beam isn't parallel); strong planar reflector on UT. Not detectable by PT/MT unless surface-breaking.
Risk: Pre-existing planar crack with a sharp tip — most standards have zero tolerance for LOF.

Lack of Penetration (LOP)

Cause: Insufficient heat or incorrect joint preparation — incomplete fusion at the weld root.
Appears: Continuous or intermittent linear indication at weld root centerline on RT; detectable by UT; visible to VT on root side if accessible.
Risk: Reduces effective throat and creates a sharp root notch — serious in cyclically loaded structures.

SLIDE 8 *Common Weld Defects — Manufacturing Discontinuities*

NOTES

Weld Defects — Cracks, Undercut & Overlap; and Service Defects

Fabrication Defects (cont.)

Cracks — always rejectable:
Hot cracks (solidification), cold/hydrogen-induced cracks (HAZ, <300°C), lamellar tearing, crater cracks. Appear as fine linear indications on RT, linear bleed-out on PT/MT, strong planar reflectors on UT. A crack is rejectable in virtually all standards regardless of size — no minimum applies.

Undercut:

Groove at the weld toe caused by excessive current or incorrect technique. Visible to VT; detectable by PT. Creates a stress concentration especially dangerous in fatigue-loaded structures. Acceptance criteria limit depth and length.

Overlap:

Weld metal rolled over base metal without fusing — low current or slow travel. Visible to VT as a lip at the toe; detectable by PT. Creates a sharp notch similar to undercut. No acceptance criteria — no overlap is acceptable.

Service-Induced Defects

Fatigue cracks:
Initiate at stress concentrations (notches, weld toes, holes, surface scratches) and propagate perpendicular to the applied cyclic stress. Often invisible to VT until large — PT, MT, or UT required for early detection.

Stress corrosion cracking (SCC):
Branching crack networks forming in susceptible alloys under the simultaneous presence of stress and a corrosive environment. Common in austenitic stainless (chlorides) and high-strength steels. Appears as branching linear PT indications.

Creep damage:
Occurs at elevated temperature under sustained stress — grain boundary cavitation progresses to cracking over time. Visible metallographically in cross-section; later-stage damage may be detectable by UT or RT.

Corrosion pitting:
Localised material loss — pits act as stress concentrations and can initiate fatigue or SCC cracks. VT reveals visible pits; UT thickness surveys map remaining wall. Severity assessed by pit depth and density.

SLIDE 9 *Weld Defects — Cracks, Undercut & Overlap; and Service Defec*

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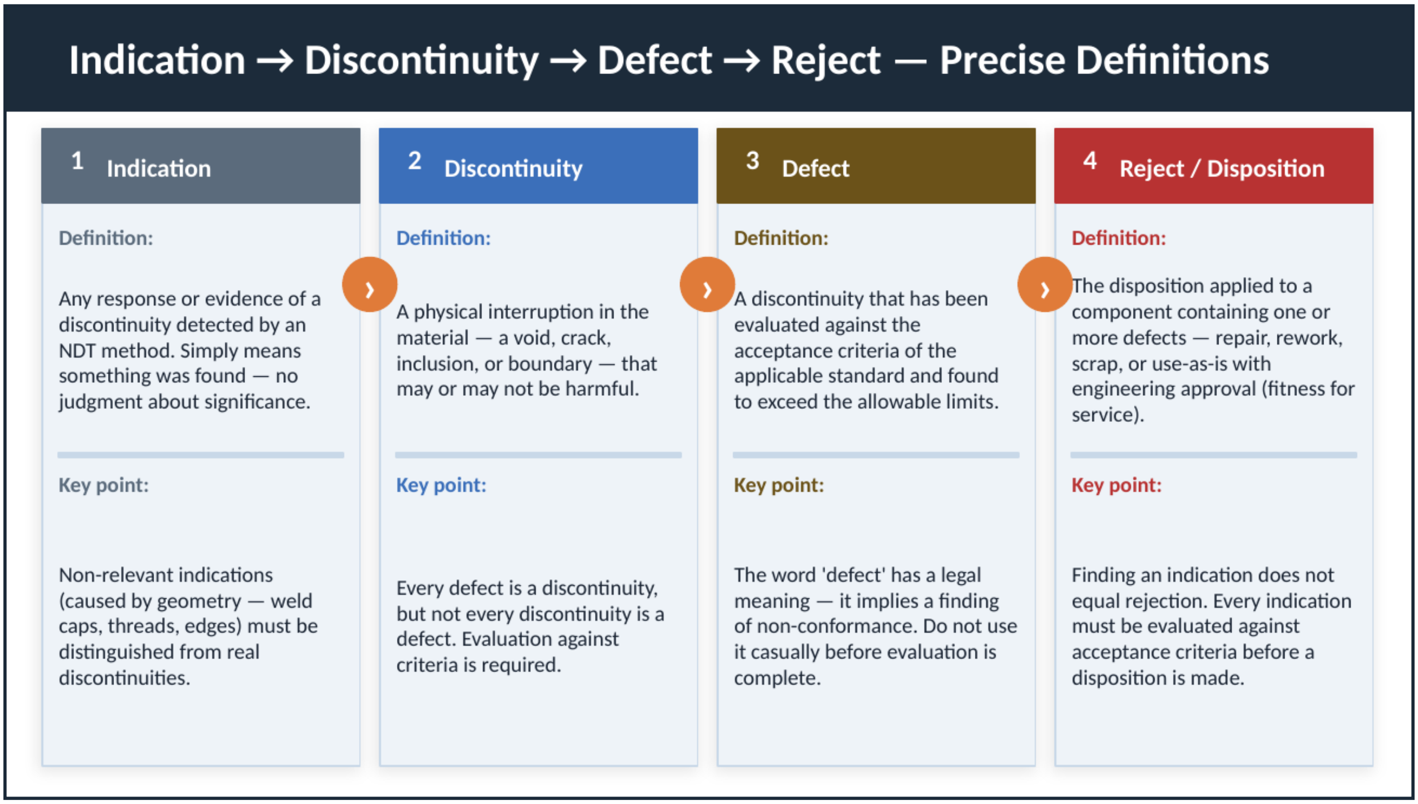

SLIDE 10 *Indication → Discontinuity → Defect → Reject — Precise Defin*

NOTES

Inspection Documentation — The Legal Record of Every Inspection

Minimum Report Content

- Component ID: drawing number, heat/lot number, weld ID or location reference
- Applicable standard and revision
- NDT method and written procedure reference number
- Equipment used: make, model, serial number, calibration due date
- Inspector name, certification level, and certification scheme
- Date, time, and location of inspection
- Environmental conditions (temperature, lighting for VT, surface condition)
- Description of any indications: location, orientation, size/length, detection method
- Acceptance criteria applied and evaluation result for each indication
- Final disposition: accept or reject, with repair or re-inspection requirements
- Inspector signature and date

Traceability & Retention

Every record must link to the physical component through unique identifiers — without traceability the record is useless in an audit or failure investigation.

In most regulated industries records must be retained for the design life of the equipment. Electronic records must be backed up and protected against alteration.

Non-Conformance Reports (NCR)

When a component is rejected, a formal NCR is raised documenting:
• Nature of the non-conformance
• Disposition: repair, rework, scrap, or use-as-is with engineering approval
• Re-inspection requirements after repair

Verbal assurances are worthless in a quality audit — if it isn't documented, it didn't happen.

SLIDE 11 *Inspection Documentation — The Legal Record of Every Inspect*

NOTES

Technician's Perspective — Documentation and Disposition in Practice

Red Flag: Report missing procedure reference and inspector certification

An inspection report arrives describing a linear PT indication within acceptance limits — but no procedure number, no equipment calibration record, and no inspector certification level are listed. Do not accept the component. Return the report, require complete documentation, and arrange re-inspection if procedural compliance cannot be demonstrated. The finding is meaningless if the inspection cannot be shown to have been performed correctly.

Caution: Indication evaluated against the wrong standard

A weld in a pressure vessel is evaluated against AWS D1.1 (structural steel) acceptance criteria. The applicable code is actually ASME Section VIII. Acceptance criteria differ between codes — a finding that passes AWS D1.1 may exceed ASME limits. This is as serious as missing a defect: confirm the governing standard before evaluating any indication.

Good Practice: Complete NCR triggers correct disposition

A root crack is found in a pipeline girth weld by UT. A Non-Conformance Report is raised immediately, the weld is marked and quarantined, and re-inspection after repair is scheduled with the same qualified procedure and inspector level. The NCR provides a traceable record of the finding, the repair, and the re-test — essential documentation if the line ever fails in service.

SLIDE 12 *Technician's Perspective — Documentation and Disposition in*

NOTES

Core Questions — Answers

What is a quality standard and why does it exist?

A standard is a document published by a recognized body (ASTM, AWS, ASME, ISO, API) defining minimum acceptable requirements for materials, processes, and inspection. Standards exist to ensure consistent quality, prevent disputes, and provide legal protection — the governing standard is the technician's rulebook for acceptance criteria, methods, and documentation.

What are the most common defects a technician will encounter?

Manufacturing: porosity (rounded voids), slag inclusions (elongated voids), lack of fusion (planar — most serious), lack of penetration (root), cracks (always rejectable), undercut, and overlap. Service: fatigue cracks, stress corrosion cracking, creep damage, and corrosion pitting. Each defect has characteristic appearance in each NDT method.

Indication vs. discontinuity vs. defect vs. reject?

Indication: something was found — no judgment yet. Discontinuity: a physical interruption — may or may not be harmful. Defect: a discontinuity that exceeds the acceptance criteria of the applicable standard. Reject: the disposition applied to a component containing a defect. Finding an indication does NOT automatically mean rejection.

What does good inspection documentation look like — and why does an incomplete report matter?

A valid report records: component ID, applicable standard, NDT procedure, equipment & calibration, inspector name & certification level, date/location, all indications with evaluation, final disposition, and signature. Missing any of these means the inspection cannot be verified — an uncertifiable inspection is no inspection at all.

SLIDE 13 *Core Questions — Answers*

NOTES

Key Takeaways

1 Standards define the rules — a technician must know which standard governs their work and apply its acceptance criteria consistently. Using the wrong standard or misapplying acceptance criteria is as serious an error as missing a defect.

2 Indication, discontinuity, defect, and reject are precise technical terms with distinct meanings — confusing them leads to incorrect dispositions and potentially unsafe components entering service.

3 Inspection documentation is not paperwork — it is the legal and technical record that proves the work was done correctly. Incomplete, inaccurate, or untraceable records undermine the entire value of the inspection process.

SLIDE 14 *Key Takeaways*

NOTES

Resources & What's Next

Week 9 Resources

Moniz Ch. 10
Materials Standards — primary reading

Quality Standards & Codes
ASTM, AWS, ASME, ISO, API reference summary

Reading Inspection Documentation
Sample MTR, inspection report walkthrough

Defect Recognition Guide
Visual identification of all common weld defects

Unit 3 Quiz
Covers Weeks 7–9 — Mechanical Testing, NDT, Standards

Week 10 — Casting

October 26, 2026 · Unit 4: Manufacturing begins

- Sand casting, investment casting, and die casting — how each process works and what it produces
- How solidification microstructure differs from wrought microstructure — and why it matters for properties
- Common casting defects — porosity, shrinkage, hot tears — and how to identify them
- Why castings require Brinell hardness testing and often need heat treatment before machining

SLIDE 15 *Resources & What's Next*

NOTES